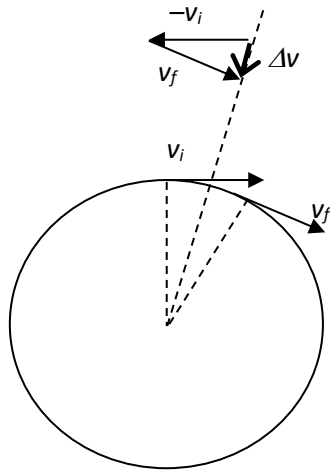


3a



- Correct vector triangle diagram showing direction change of velocity towards centre of circle

B1

(Learning points:

- Velocity v_i represents the object's velocity at one instant and v_f its velocity at a later instant.
- In the vector triangle, the change in velocity Δv ($=v_f - v_i$), is due to a change in the velocity's direction, while the magnitude of v_i and v_f (length of arrows) are the same)

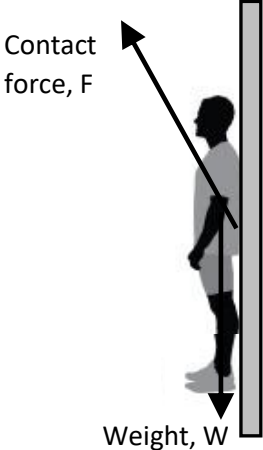
- Explanation relating change in direction of velocity resulting in acceleration:

While magnitude of velocity remains the same, the direction of velocity changes resulting in an acceleration.

B1

(Learning points:

- Direction of acceleration (or force) is towards centre of circle, same as the direction of the change in velocity. Since acceleration is perpendicular to velocity, speed remains constant.)

bi	Correct direction of forces and labelled clearly - gravitational force by earth on man or weight vertically downwards - contact force by wall on man (diagonal to the upper left) showing Vertical component of contact force about equal in magnitude to weight OR Contact force represented by: Friction vertically upwards about equal magnitude to weight and Normal Reaction perpendicular to wall 	B1 B1
bii	The <u>upward frictional force is equal and opposite to the downward gravitational force</u> acting on the rider. Hence, the rider would not fall off. The <u>normal contact force by the wall on the rider provides for the centripetal force</u> necessary for him to move together with the barrel in a circular path.	B1 B1
biii	acceleration = $r\omega^2 = r(2\pi f)^2 = 2.5(2\pi)^2(33/60)^2$ = 29.9 m s⁻²	C1 A1
biv	As the <u>net force acting on the man</u> (normal contact force) is <u>always perpendicular to the velocity (or direction of motion) of the man</u> at any instant. so work done on the rider is zero OR Since speed is constant, there is no change in kinetic energy of the rider, so work done on the rider is zero <i>Learning point:</i> <i>As the instantaneous velocity is in the same direction of instantaneous displacement, hence, work done which is the product of force and the displacement in the direction of the force would be zero.</i>	M1 A1 [M1] [A1]
	Total:	10

